Dharohar Atlas

Digital Heritage Preservation & Spatial Intelligence Platform

SYSTEM ARCHITECTURE DOSSIER

Repository: `github.com/ramng-rgb/DHAROHAR-ATLAS`

Platform Baseline: PostGIS • Next.js 14 • Gemini AI • Three.js

1. Problem Solved

India's ancient monuments, subterranean stepwells (*vavs/baolis*), rock-cut sanctuaries, and epigraphic folios face catastrophic preservation risks, academic isolation, and digital fragmentation. Traditional heritage archives suffer from four fundamental failures:

- **Information Siloing & Spatial Disconnection:** Geospatial coordinates, architectural metadata, historical dynasties, and folk narratives exist in disconnected spreadsheets or isolated archaeological surveys lacking spatial GIS topology.
- **Fragile Physical Epigraphy & Inaccessible Transcripts:** Thousands of ancient stone inscriptions and palm-leaf manuscripts (Devanagari, Sanskrit, Old Gujarati) deteriorate physically without high-accuracy OCR or accessible multi-language transcription pipelines.
- **Spatial Query Bottlenecks:** Conventional web maps crash or lag when rendering dense regional historical clusters without database-level spatial indexing or adaptive bounding-box viewport clustering.
- **AI Hallucination in Cultural Heritage:** General LLMs routinely fabricate historical dates, dynasties, and folklore legends without grounded attribution or cited primary source grounding.

> **Core Solution: Dharohar Atlas** bridges this divide as an open-access, full-stack digital preservation engine. It combines PostGIS-backed spatial clustering across 101 verified monuments, sub-millimeter 3D photogrammetric twin streaming, an asynchronous OCR/translation pipeline for Devanagari epigraphy with human-in-the-loop review, and a grounded hybrid vector/BM25 retrieval engine resistant to hallucinations.

2. Solution & System Architecture

The platform follows a decoupled, resilient 4-tier microservice architecture designed for ultra-low latency interactive exploration and high-throughput archival processing:

TIER 1: WEB CLIENT

- Next.js 14 App Router & React 18
- MapLibre GL 6 Vector Engine
- Three.js / Drei Photogrammetry
- Zustand State & URL Sync
- Framer Motion Fluid Transitions

TIER 2: API GATEWAY

- Python 3.12 FastAPI Framework
- Pydantic v2 Strict Input Schemas
- Tiered Rate Limiter (Anon / Auth)
- ETag Caching & GZip / Brotli
- Circuit Breaker Resilience

TIER 3: SPATIAL & VECTOR

- PostgreSQL 16 + PostGIS Engine
- pgvector (768-dim Embeddings)
- pg_trgm Trigram / Full-Text
- Redis 7 Distributed Cache
- Reciprocal Rank Fusion (RRF)

TIER 4: ARCHIVAL & AI

- PyTesseract & Google Gemini Vision
- Neural Translation Engine
- Asynchronous Celery / Redis Workers
- Confidence Scoring & Line BBoxes
- Audit Logging & Versioning

Request Lifecycle: Viewport queries from MapLibre trigger spatial ST_MakeEnvelope bounding-box queries in PostGIS. Manuscript uploads stream to asynchronous background workers that slice PDF folios, run multi-engine OCR, and calculate SacreBLEU / EditDistance confidence before committing to pgvector.

3. Relational, Geospatial & Vector Database Schema

The relational model enforces strict referential integrity, spatial indexing, and soft-delete capabilities managed through 8 Alembic migration revisions:

TABLE NAME	KEY ATTRIBUTES & DATA TYPES	INDEXES & CONSTRAINTS
sites	`id (UUID)`, `slug (VARCHAR)`, `name (VARCHAR)`, `state (VARCHAR)`, `site_type (ENUM)`, `location (GEOGRAPHY Point 4326)`, `period_start_year`, `period_end_year`, `hero_image (VARCHAR)`, `hero_image_attribution (TEXT)`, `search_vector (TSVECTOR)`	Spatial GIST index on `location`, GIN index on `search_vector`, Check constraints on latitude [-90, 90], longitude [-180, 180], and period chronology.
site_media	`id (UUID)`, `site_id (FK)`, `kind (ENUM)`, `remote_url`, `licence_short`, `licence_url`, `author (TEXT)`, `credit (TEXT)`, `is_placeholder (BOOL)`, `verified (BOOL)`	Foreign Key with cascade delete to `sites.id`. Indexed on `(site_id, kind, sort_order)`. Mandatory license verification filter.
archival_documents	`id (UUID)`, `site_id (FK)`, `title`, `language`, `script`, `source_archive`, `folio_number`, `storage_path`, `status (ENUM: pending/processing/ready/failed)`	B-Tree index on `(site_id, status)`. Audit timestamping on creation and state transitions.
document_lines	`id (UUID)`, `document_id (FK)`, `line_number`, `bounding_box (JSON)`, `transcription (TEXT)`, `translation (TEXT)`, `ocr_confidence (FLOAT)`, `is_corrected (BOOL)`	Foreign Key to `archival_documents.id`. Supports line-level bounding box coordinate rendering and optimistic human corrections.
search_chunks	`id (UUID)`, `source_type (site/document)`, `content_text (TEXT)`, `embedding (vector(768))`, `tsv (TSVECTOR)`, `chunk_metadata (JSONB)`	HNSW vector cosine distance index (`vector_cosine_ops`) for sub-5ms semantic similarity, combined with GIN index on `tsv` for RRF hybrid search.
curated_routes	`id (UUID)`, `title`, `slug`, `theme`, `estimated_days`, `stops (One-to-Many SiteRouteStop)`, `polyline_geojson (JSON)`	Ordered waypoint sequencing with Haversine distance aggregation and audio tour step synchronization.

4. Complete Technology Stack

FRONTEND WEB ARCHITECTURE

Next.js 14 App Router, React 18, TypeScript, Vanilla Tailwind CSS, MapLibre GL 6 (Vector Tiles & GeoJSON), Three.js / React Three Fiber / Drei, Lucide React, Zustand State Management.

BACKEND & API SERVICES

Python 3.12, FastAPI (ASGI), Pydantic v2 (Strict Data Models), SQLAlchemy 2.0 (asyncpg Async Engine), Alembic (Schema Migrations), Redis 7 (Broker & Distributed Rate Limiting).

DATABASE & SPATIAL ENGINES

PostgreSQL 16 with PostGIS 3.4 (Spatial Geometry/Geography), `pgvector` (768-dim embeddings), `pg_trgm` (Trigram fuzzy indexing), `unaccent` for transliterated search.

AI / ML & VISION PIPELINE

Google Gemini 2.0 Flash & `text-embedding-004` via official Google GenAI SDK, PyTesseract, SacreBLEU, EditDistance, NumPy, Pillow, PyPDFium2, Resilient Fallback Heuristics.

QUALITY & INFRASTRUCTURE

Docker Multi-stage Builds, Docker Compose, Pytest (92 unit/integration tests), Vitest (42 frontend tests), Playwright E2E, Ruff Linter, Mypy Static Typechecker.

LICENSING & COMPLIANCE

Strict MediaWiki API integration filtering exclusively for CC-BY, CC-BY-SA, CC0, and Public Domain. Automated image attribution ledger and CC credit export.

5. Value Proposition & Potential Scalability

National Cultural & Academic Value:

- **Universal Open Access:** Unlocks democratized access to 101 authentic Indian heritage monuments, from celebrated UNESCO World Heritage stepwells (Rani ki Vav) to remote desert forts.
- **Epigraphic Revitalization:** Converts degraded stone inscriptions and rare folios into searchable, machine-readable text with line-aligned English translations.
- **Ethical Cultural Tourism:** Waypoint-guided thematic journeys empower local tourism and scholarly research without commercial paywalls.
- **Zero Licensing Infringements:** Strict automated attribution ledger respects artists and archives via verified Wikimedia Commons CC licenses.

Horizontal & Vertical Scalability:

- **Geospatial Partitioning:** PostGIS spatial indexes (GIST) partition points across geohashes, sustaining 50,000+ simultaneous monuments without bounding-box lag.
- **Sub-Millisecond Read Latency:** Redis layer caches frequent GeoJSON cluster tiles and route payloads with HTTP ETag 304 Not Modified validation.
- **Decoupled Pipeline Workers:** Archival OCR workers scale horizontally in independent Docker containers across Celery/Redis task queues.
- **Serverless Model Delivery:** 3D photogrammetry assets and optimized WebP thumbnails stream statically via Edge CDNs without backend load.

6. Current Limitations & Strategic Roadmap

Current Technical Limitations:

- **Regional Script OCR Baselines:** Sharada, Grantha, and Brahmi scripts currently require fine-tuned heuristic fallbacks due to limited open-source OCR training datasets.
- **Mobile 3D WebGL Constraints:** Dense photogrammetric meshes (>50MB) require downsampling to GLTF LODs on low-memory mobile browser environments.
- **Offline Map Caching:** Full offline tile storage currently requires Progressive Web App (PWA) service worker installation.

Development & Expansion Roadmap:

- **Phase 1 (Q1 2027) — Crowdsourced Epigraphy:** Launch community peer-review portal enabling certified epigraphists to annotate and verify manuscript transcriptions with Git-style versioning.
- **Phase 2 (Q2 2027) — Audio Guide Synthesis:** Multilingual neural text-to-speech generating regional Indian audio guides (Hindi, Gujarati, Tamil, Bengali).
- **Phase 3 (Q3 2027) — WebXR Spatial Walkthroughs:** Immersive VR/AR walkthroughs of subterranean stepwells directly in Apple Vision Pro & Meta Quest browsers.
- **Phase 4 (Q4 2027) — ASI Ingestion Pipeline:** Native integration with Archaeological Survey of India excavation records and satellite radar data.